

Satellite & Space Networking

A Research Report

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TABLE OF CONTENTS

Sr. No.	Contents	Page No.
1	Introduction to Satellite & Space Networking	3
2	Types of Satellites	4
3	Satellite Communication Process	5
4	Advantages of Satellite Networking	6
5	Disadvantages of Satellite Networking	7

1. Introduction to Satellite & Space Networking

Satellite & Space Networking refers to communication systems that use satellites in space to transmit data, voice, and video signals across long distances. These networks are important for global communication, especially in areas where traditional networks like cables or towers are not available.

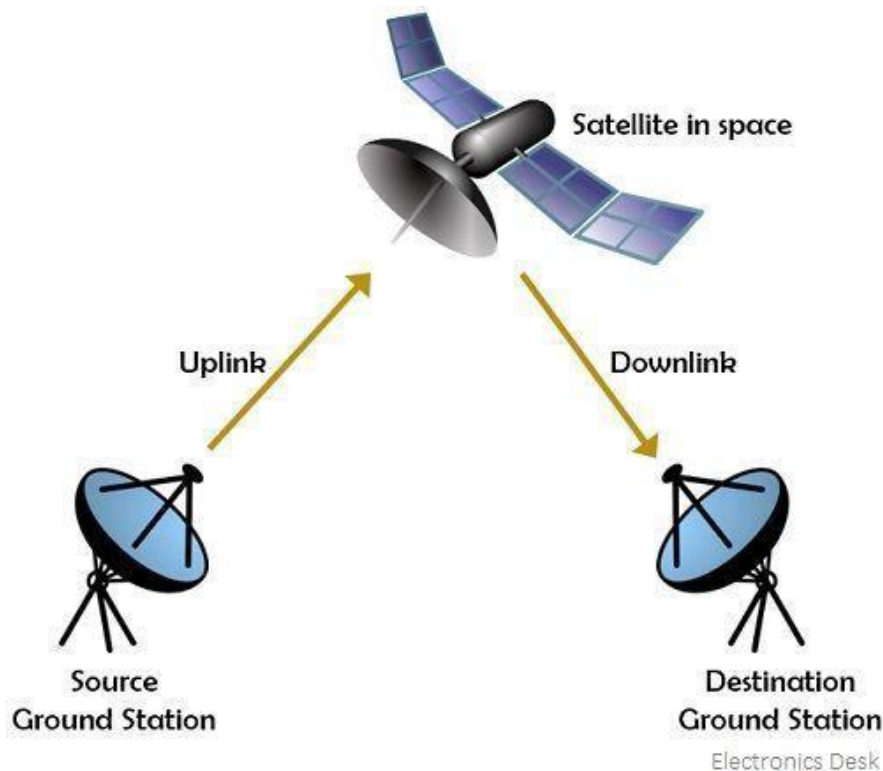
Satellites act as relay stations in space. They receive signals from one location on Earth (uplink) and transmit them back to another location (downlink). This allows communication over very large distances, even between different continents.

Key Features:

- Provides **global coverage**
- Works in **remote and rural areas**
- Supports **long-distance communication**
- Used for broadcasting and internet services

Basic Working:

1. Signal is sent from Earth station (**uplink**)
2. Satellite receives and processes signal
3. Signal is sent back to Earth (**downlink**)



2. Types of Satellites

Satellites used in networking are classified based on their **orbit around the Earth**. Each type has different altitude, speed, and use.

1. LEO (Low Earth Orbit)

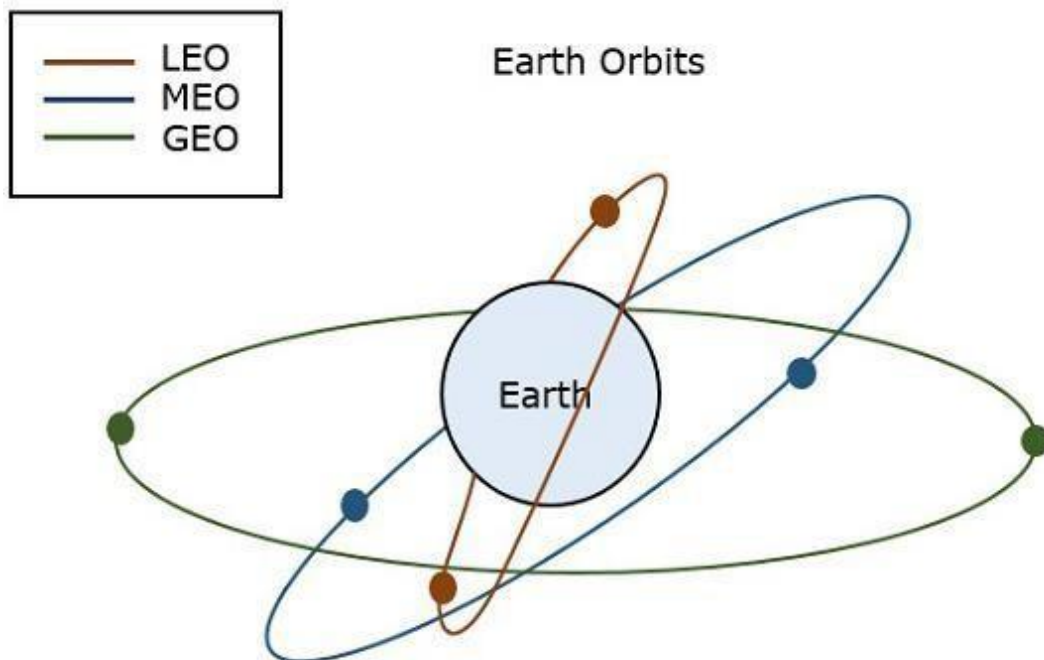
- Altitude: **160 – 2,000 km**
- Moves very fast around Earth
- Low delay (latency)
- Used for: Internet services (e.g., Starlink)

2. MEO (Medium Earth Orbit)

- Altitude: **2,000 – 35,786 km**
- Moderate speed and delay
- Used for: GPS navigation systems

3. GEO (Geostationary Orbit)

- Altitude: **35,786 km**
- Remains fixed over one location
- High delay but wide coverage
- Used for: TV broadcasting, weather monitoring



3. Satellite Communication Process

The **Satellite Communication Process** explains how data is transmitted from one place on Earth to another using a satellite.

Steps Involved:

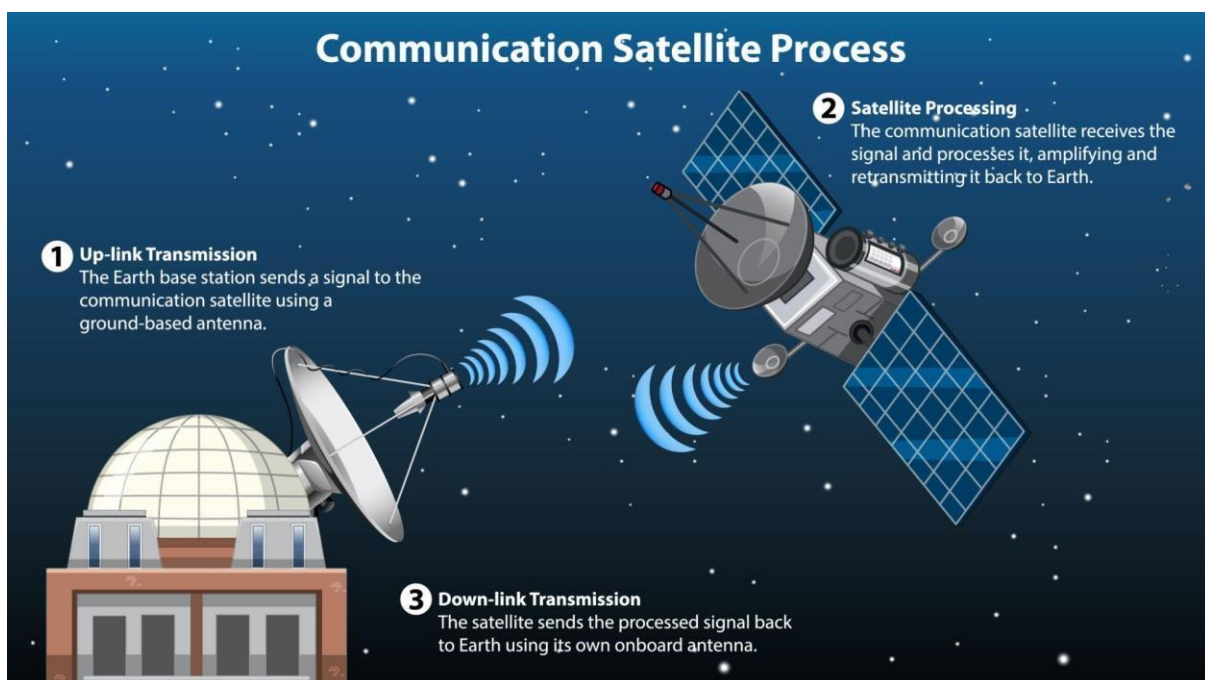
1. **Uplink (Sending Signal)**
The Earth station sends a signal to the satellite using radio waves.
2. **Signal Processing (Transponder)**
The satellite receives the signal and amplifies or modifies it using a **transponder**.
3. **Downlink (Receiving Signal)**
The satellite sends the signal back to another Earth station.

Key Components:

- **Earth Station** – Sends and receives signals
- **Satellite** – Acts as a relay in space
- **Transponder** – Processes the signal

Important Points:

- Communication uses **microwave frequencies**
- Covers large geographical areas
- Used for both one-way and two-way communication



4. Advantages of Satellite Networking

Satellite Networking provides many benefits, especially for global and remote communication.

Key Advantages:

- **Global Coverage**
Can connect any part of the world, including oceans and remote areas.
- **Works in Remote Areas**
No need for cables or towers, ideal for rural and inaccessible locations.
- **Broadcasting Capability**
Supports TV, radio, and live streaming to large audiences.
- **Reliable Communication**
Useful during disasters when other networks fail.

Why It Is Useful:

- Connects places where internet is not available
- Supports navigation (GPS)
- Helps in emergency communication



5. Disadvantages of Satellite Networking

Although **Satellite Networking** has many benefits, it also has some limitations.

Key Disadvantages:

- **High Latency**
Signals travel long distances, causing delay (especially in GEO satellites).
- **Weather Interference**
Rain, storms, and clouds can weaken signals (rain fade).
- **High Cost**
Launching and maintaining satellites is very expensive.
- **Limited Bandwidth**
Data capacity is limited compared to fiber networks.
- **Signal Loss**
Signals may weaken over long distances.

Why It Matters:

These issues can affect communication quality, speed, and reliability.

Conclusion:

Despite its global reach, satellite networking faces challenges like delay, cost, and environmental interference.

